

MobileWAM: Bridging World Action Models to Mobile Manipulation with Chain-of-Foresight

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World action models (WAMs) built on video generation backbones are a rising recipe for robot learning, yet remain confined to tabletop manipulation. Mobile manipulation demands simultaneous locomotion and whole-body manipulation amid scene-scale dynamics, yet is still dominated by dynamics-blind visual encoders with hand-crafted coordination. We bridge this gap with MobileWAM, a mixture-of-transformers architecture that fuses a pretrained video diffusion transformer with a lightweight action expert through layerwise joint attention, translating internet-scale motion priors into whole-body control. To reconcile the heterogeneous dynamics of moving and manipulating, each feed-forward layer of the action expert becomes a three-expert mixture of shared, locomotion, and manipulation experts, softly routed by the motion intent in the action tokens. To densify supervision, we further propose Chain-of-Foresight (CoF): intermediate representations sequentially predict a chain of future latent chunks, each step conditioned on its predecessor. CoF pairs naturally with our decoupled video-action denoising scheme. At deployment, the WAM serves as a pure current-frame encoder; foresight acts only through gradients, so at inference the foresight chain and video generation are discarded, leaving only policy-level cost. MobileWAM surpasses state-of-the-art mobile manipulation policies on ManiSkill-HAB and fine-tunes to a real ARX Lift2 mobile manipulator across diverse tasks with strong generalization. Code will be released upon acceptance.

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1 Introduction

Is mobile manipulation merely tabletop manipulation with a mobile base, or “navigation plus manipulation” stapled together? We argue no: coupling locomotion with manipulation is multiplicative rather than additive. Three gaps separate the two settings. *Viewpoint*: tabletop cameras are fixed; mobile robots’ cameras sweep across places, demanding perception consistent under relentless ego-motion. *Multimodality*: with the base free, many base-arm paths reach the same goal, making the action distribution far more multimodal (Lim et al., 2026). *Causality*: an early navigation error silently dooms a later grasp. Mobile manipulation is long-horizon not by frame count, but by causal depth and spatial variation.

Tabletop tasks often reduce to vision-to-action mappings where short-horizon VLAs excel (Brohan et al., 2023; Kim et al., 2024). These shallow mappings evaporate in mobile manipulation, where representations must encode how the world *will* evolve.

Video-generative diffusion transformers address this need: pretrained on internet-scale video, they internalize physical priors and synthesize coherent futures, fueling world action models (WAMs) (Bi et al., 2026; Li et al., 2026; Yuan et al., 2026). Yet successes concentrate on fixed-base manipulation; mobile manipulation still relies on dynamics-blind encoders: point-cloud fusion with two-stage conditioning (Chen et al., 2025b), or decoders supervised with privileged GT segmentation (Tu et al., 2026; Lin et al., 2026). The natural marriage has yet to be consummated.

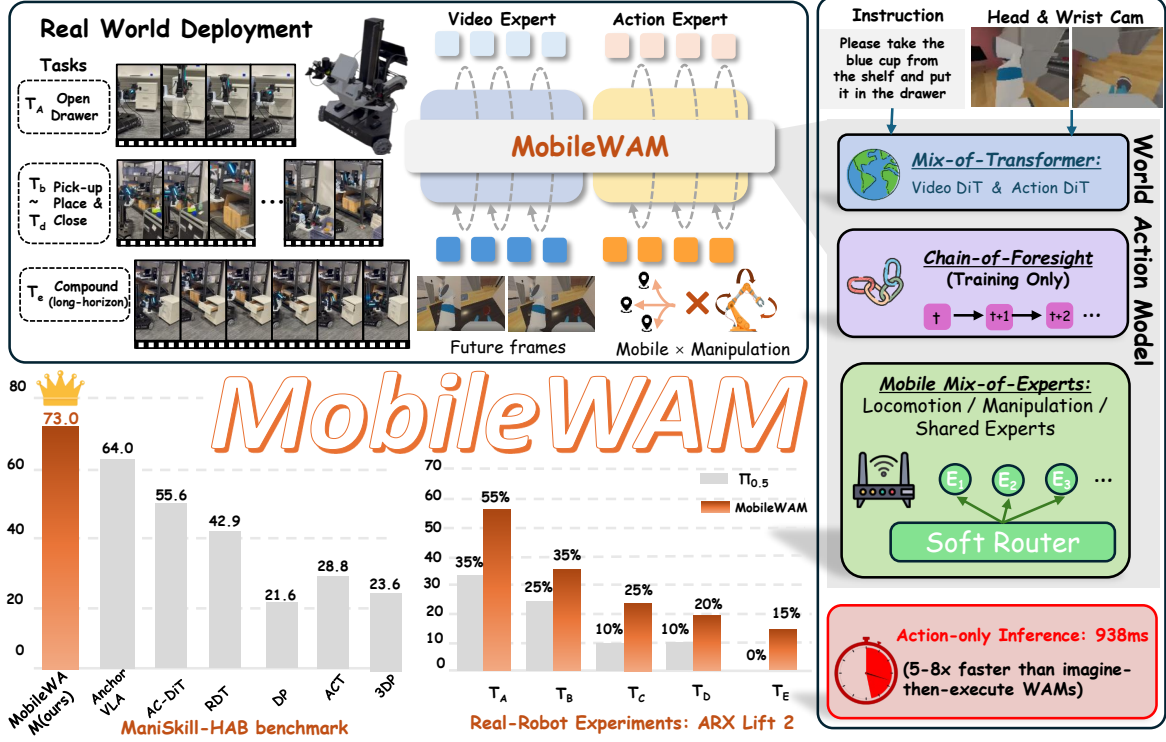


Figure 1 MobileWAM at a glance. *Top-left:* five real-robot tasks of increasing horizon (T_A – T_E) on an ARX Lift2. *Top-center/right:* Mixture-of-Transformers fuses video and action experts; Chain-of-Foresight (training only) chains future-latent predictions via an RNN-style belief; Mobile MoE routes locomotion, manipulation, and shared experts via a soft router. *Bottom-left:* ManiSkill-HAB mean S.R. 73.0%, leading all listed baselines. *Bottom-right:* real-robot success rates vs. $\pi_{0.5}$ across T_A – T_E ; action-only inference at 938 ms is 5–8 $\times$ faster than generate-then-act WAMs.

We take an early step with MobileWAM (Figure 1), a WAM systematically adapted to mobile manipulation. It grafts a lightweight action expert onto a pretrained video diffusion transformer via layerwise joint attention, mixture-of-transformers style (Liang et al., 2025), steeping whole-body action denoising in the backbone’s motion priors at every layer. Two designs tailor it further. First, locomotion and manipulation are distinct regimes; forcing one pathway to serve both causes destructive interference. Our *mobile mixture-of-experts* routes each feed-forward layer across three experts (shared, locomotion, manipulation) softly routed by motion intent. Second, inspired by next-latent prediction (Teoh et al., 2025) in LLMs, we propose *Chain-of-Foresight* (CoF) that embeds a latent dynamics model: an RNN-style chain where step k denoises the k -th future latent and passes its belief to step $k+1$ with hidden states, which carry the world’s physical state forward. Following the decoupled denoising paradigm (Yuan et al., 2026), deployed MobileWAM acts as a *current-frame encoder* that caches one backbone pass and denoises actions against it. CoF is deleted at inference, yielding 5 $\times$ –8 $\times$ speedups over imagine-then-execute WAMs.

MobileWAM surpasses state-of-the-art on ManiSkill-HAB (Shukla et al., 2025) with RGB-only inputs and one-stage training, and transfers to a real ARX Lift2, beating $\pi_{0.5}$ (Black et al., 2025) with margins that grow with task horizon.

Our contributions are fourfold:

- **Bridging WAMs to mobile manipulation.** To our knowledge, we present the first systematic adaptation of video-generation WAMs to whole-body mobile manipulation, achieving state-of-the-art ManiSkill-HAB results with RGB-only inputs and one-stage training.
- **Chain-of-Foresight.** A recurrent, serially chained future-latent prediction objective that densifies temporal supervision and strengthens long-range spatio-temporal and causal consistency, at zero inference cost.
- **Mobile MoE.** A three-expert mixture (shared, locomotion, manipulation) inside the action expert, routed by motion intent, reconciling the heterogeneous dynamics of moving and manipulating.

- **Real-robot deployment.** MobileWAM fine-tunes to the ARX Lift2 and consistently outperforms $\pi_{0.5}$ across five household tasks, with the largest gains on the longest horizons.

2 Related Work

2.1 World Action Models

World action models couple a generative observation model with an action head: imagining motion consequences yields physically grounded representations. Early systems were two-stage: UniPi inverts a generated video into actions (Du et al., 2023); VPP and Seer condition policies on predicted states (Hu et al., 2025b; Tian et al., 2025). Recent WAMs collapse stages into end-to-end pretrained diffusion transformers: Motus (Bi et al., 2026), LingBot-VA (Li et al., 2026), Fast-WAM (Yuan et al., 2026), GigaWorld-Policy (GigaWorld Team, 2026); ImageWAM even finds an image-editing backbone sufficient for tabletop WAMs (Zhang et al., 2026), suggesting fixed-base manipulation barely needs long-horizon video prediction. S-VAM pushes the inference-foresight trade-off further by self-distilling multi-step generative priors into a single forward pass (Yan et al., 2026). Yet all address fixed-base tasks, blunting their core advantage. MobileWAM adopts this line’s mixture-of-transformers fusion and decoupled denoising, but extends it to where the camera rides on the robot base.

2.2 Mobile Manipulation

Mobile manipulation research splits into two camps. The *modular* camp decouples navigation from manipulation via trajectory optimization (Wu et al., 2024; Yan et al., 2025), base-pose optimization (Yang et al., 2025; Wu et al., 2025a,b), or LLM/VLM planners (Wang et al., 2025; Chen et al., 2025a), at the price of hand-designed interfaces that leak errors across stages. The *end-to-end* camp learns whole-body control directly (Fu et al., 2024; Jiang et al., 2025; Li et al., 2025; Zhu et al., 2026): dense policies (Su et al., 2025), anchored diffusion (Lim et al., 2026), memory-augmented VLAs (Lin et al., 2025), reasoning-augmented VLAs that inject visual-linguistic chains of thought (Zhong et al., 2026), and π_0 -family models dominating BEHAVIOR (Black et al., 2024, 2025). Closest to ours: AC-DiT requires 3D point clouds and two-stage training (Chen et al., 2025b); SG-VLA uses auxiliary decoders with privileged GT segmentation (Tu et al., 2026); WEM separates world and ego streams but presupposes that separation (Lin et al., 2026); ABot-M0.5 explores WAMs with latent actions (Chen et al., 2026). MobileWAM needs none of the above. Its advantage runs deeper than a leaner recipe: the consistency others engineer explicitly is inherited from video pretraining and reinforced from within. CoF turns intermediate representations into an RNN-style latent dynamics model that directly supervises physical regularity, while the mobile MoE lets locomotion and manipulation experts specialize without severing the shared structure that binds them.

2.3 Future Prediction as Auxiliary Supervision

A complementary line supervises *representations* with the future rather than generating pixels: latent-dynamics planning over frozen features (Zhou et al., 2025), foresight-conditioned inverse dynamics (Tian et al., 2025), and reconstruction-based perception for VLAs (Song et al., 2025). Most related, the Belief State Transformer and Next-Latent Prediction show that predicting one’s *own future latent state* injects a recurrent inductive bias, compressing history into belief states (Hu et al., 2025a; Teoh et al., 2025). CoF transplants this into WAMs with two twists: targets are *future video latents* grounded in physical scene evolution, and the chain is serially unrolled so each step conditions on its predecessor’s belief. The chain is never consulted at inference; it exists solely to shape the representations the action expert reads.

3 Method

3.1 Overview

At time t the robot receives an observation $o_t = \{I_t^{\text{head}}, I_t^{\text{wrist}}\}$ from head and wrist RGB cameras, proprioceptive state $s_t \in \mathbb{R}^{d_s}$ (joint positions, velocities, and base odometry), and language instruction ℓ . The policy outputs

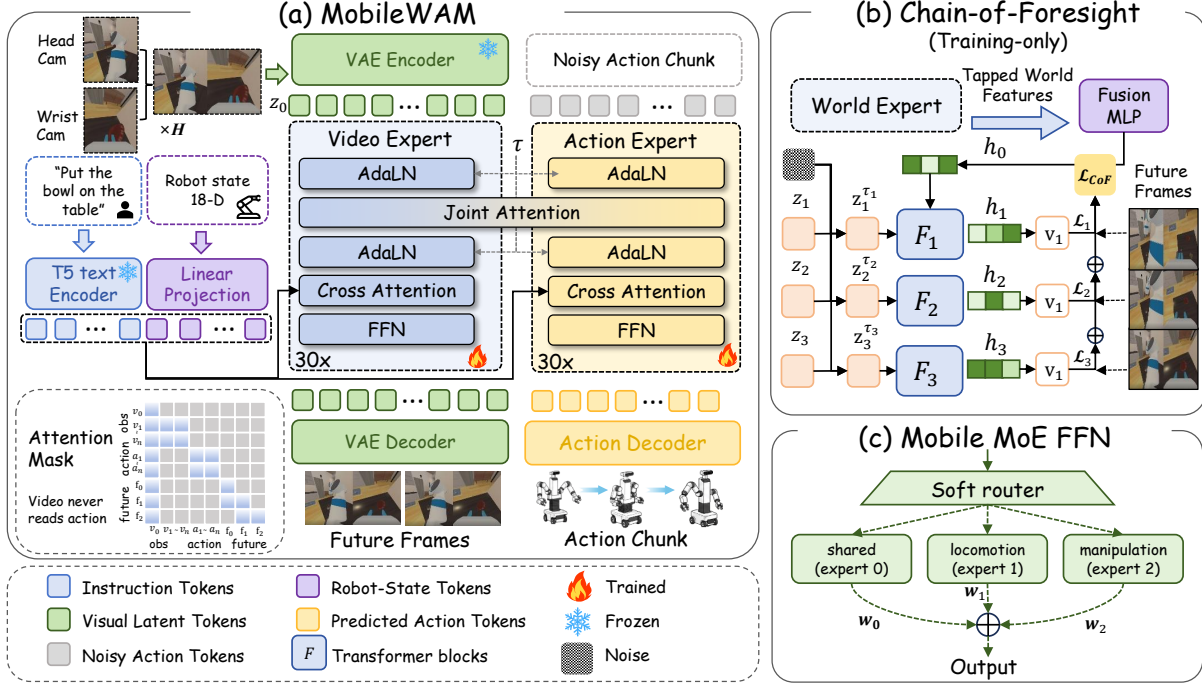


Figure 2 MobileWAM architecture. **(a) Main model.** The world and action experts fuse via asymmetric layerwise joint attention (actions read visuals; visuals never read actions); language and proprioception are injected via per-block cross-attention; every feed-forward layer of the action expert is a three-expert mobile MoE (shared $\oplus$ locomotion $\oplus$ manipulation) softly routed by motion-intent embeddings. **(b) Chain-of-Foresight.** Four uniformly spaced backbone layers fuse into belief h_0 ; depth-specific F_1, F_2, F_3 (separate weights, init. from later backbone blocks) chain over f_0, f_1, f_2 , each attending to the previous belief and current observation. Weights $w_1 > w_2 > w_3$ discount farther futures. The CoF branch is discarded at inference. **(c) Attention mask.** Token groups: v_0 (current observation, clean latent); $v_1 \sim v_n$ (future video latents, noisy, bidirectional); $a_1 \sim a_n$ (noisy action chunks); $f_0 \sim f_{K-1}$ (foresight latents, training only). $v_1 \sim v_n$ attend to v_0 and each other; actions attend to v_0 only; each foresight step f_k ($k \geq 1$) attends to v_0 and f_{k-1} , while f_0 attends to v_0 and itself. Video and action denoising are fully decoupled, enabling current-frame-encoder inference.

a whole-body action chunk $\mathbf{a}_{t:t+H} = (\mathbf{a}_t, \dots, \mathbf{a}_{t+H-1})$ with each $\mathbf{a} \in \mathbb{R}^{d_a}$; $d_a=13$ dimensions jointly command the arm (7 joints), gripper, head pan/tilt, torso lift, and base linear/angular velocities. During training the model also observes a future video segment $V_{t:t+T}$ of horizon T and learns

$$p_{\theta}(\mathbf{a}_{t:t+H}, V_{t:t+T} \mid o_t, \mathbf{s}_t, \ell); \quad (1)$$

at deployment only the action marginal is sampled. Both modalities are trained with flow matching, detailed below.

MobileWAM (Figure 2) couples a large *world expert* (pretrained video diffusion transformer) with a lightweight *action expert* via layerwise joint attention. An asymmetric attention mask decouples video and action denoising so that deployment can drop the video branch without changing the current-frame features the policy reads. Chain-of-Foresight (CoF) provides a training-only recurrent supervision signal at zero inference cost.

3.2 World Expert

The backbone is a pretrained text-and-image-to-video diffusion transformer (Wan Team, 2025; Peebles and Xie, 2023) ($L=30$ blocks, width 3072). A frozen 3D VAE compresses video $16\times$ spatially and $4\times$ temporally; one latent tick covers one action chunk ($H=4$), aligning visual and motor clocks. Both camera views are composited side by side, enforcing cross-view consistency via spatial attention. The current frame enters as a clean latent (timestep zero); subsequent frames carry noise, so the backbone denoises the future *given the*

present. Language ℓ is encoded by a frozen T5-family encoder (Raffel et al., 2020) and injected via per-block cross-attention.

3.3 Action Expert and Mobile MoE

A lightweight action-denoising transformer (width 1024) runs in lockstep: at each of the 30 layers, its action tokens join the backbone’s visual tokens in shared self-attention, mixture-of-transformers style (Liang et al., 2025; Bi et al., 2026), with separate projections but a concatenated sequence. Noisy action chunks are linearly embedded as action tokens; $\mathbf{s}_t$ is appended to the text context for cross-attention.

Locomotion and manipulation are distinct motion regimes: base commands navigate through free space; arm and gripper commands resolve precise contact. Forcing a single pathway to handle both causes destructive interference: updates that sharpen grasping compete with those shaping approach trajectories. We replace every feed-forward layer of the action expert with a three-expert mixture: a *shared* expert for regime-agnostic structure, plus dedicated *locomotion* and *manipulation* experts (Shazeer et al., 2017). A router reads the mean-pooled noisy action embedding and produces soft weights via temperature-scaled softmax. Experts are cloned from the dense layer; the router is zero-initialized, so specialization emerges only where data demands it. No load-balancing loss is needed. The design echoes the disentanglement in Chen et al. (2026), but as a drop-in, single-stage module.

3.4 Chain-of-Foresight

A single prediction chunk asks “what happens next?” Mobile manipulation demands “and then? and then?” CoF converts this into an auxiliary objective (Figure 2b).

Let $\mathbf{H}^{(l)}$ be the backbone’s hidden states at layer l for current-observation tokens. We tap four uniformly spaced layers $\{4, 12, 20, 30\}$ and fuse them via a two-layer MLP g :

$$\mathbf{h}_0 = g\left(\left[\mathbf{H}^{(4)}; \mathbf{H}^{(12)}; \mathbf{H}^{(20)}; \mathbf{H}^{(30)}\right]\right). \quad (2)$$

Shallow layers carry geometry; deep layers carry semantics; either extreme or all-30 concatenation degrades the chain (Table 4).

Depth-specific modules $F_1, \dots, F_K$ (separate weights, initialized from later backbone blocks) are chained over future latents $\mathbf{z}_1, \dots, \mathbf{z}_K$ ($f_0, \dots, f_{K-1}$ in Figure 2). At step k , F_k takes belief $\mathbf{h}_{k-1}$ and noised target $\mathbf{z}_k^{\tau_k}$, cross-attends to $(\ell, \mathbf{s}_t, o_t)$ under a block-causal mask, and outputs a velocity estimate and foresight belief:

$$(\hat{\mathbf{v}}_k, \mathbf{h}_k) = F_k(\mathbf{h}_{k-1}, \mathbf{z}_k^{\tau_k}; \ell, \mathbf{s}_t, o_t), \quad k = 1, \dots, K. \quad (3)$$

The belief is the sole conduit between steps, forcing $\mathbf{h}_k$ to summarize the evolving world: the latent analogue of belief-state learning (Hu et al., 2025a; Teoh et al., 2025).

3.5 Attention Mask and CoF Interaction

The joint attention mask is asymmetric (Figure 2): current tokens attend only among themselves; future video tokens attend to all visual tokens; action tokens attend to current tokens only; no video token sees any action token. This ensures the current-observation representation is identical at training and deployment.

At inference, the WAM becomes a *current-frame encoder*: per-layer KV pairs are cached from one backbone pass; action tokens are iteratively denoised against this cache without instantiating any future frame (Yuan et al., 2026). CoF is fully compatible: its gradients flow from each foresight step through the fusion MLP into the tapped backbone layers, and from there into the joint attention that action tokens read, pressuring the backbone to encode scene dynamics in its current-observation representation. The foresight module and fusion MLP are deleted at inference, adding zero parameters or FLOPs.

3.6 Losses

Both video and action are trained with flow matching (Lipman et al., 2023). For a clean sample $\mathbf{x}$ (video latent or action chunk), a noisy interpolant $\mathbf{x}^\tau = (1-\tau)\mathbf{x} + \tau\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, is formed at noise level $\tau \in [0, 1]$; the network predicts velocity field $\hat{\mathbf{v}}$ approximating $\mathbf{v} = \boldsymbol{\varepsilon} - \mathbf{x}$:

$$\mathcal{L}_v = \mathbb{E} \|\hat{\mathbf{v}}_{\text{vid}} - \mathbf{v}_{\text{vid}}\|^2, \quad \mathcal{L}_a = \mathbb{E} \|\hat{\mathbf{v}}_{\text{act}} - \mathbf{v}_{\text{act}}\|^2, \quad (4)$$

with independent, shift-skewed timestep sampling so either branch can be denoised alone. Sharing one formalism lets action tokens attend to visual tokens under the same noise semantics, making layerwise fusion coherent.

CoF contributes a depth-decayed term, discounting farther futures:

$$\mathcal{L}_{\text{CoF}} = \sum_{k=1}^K w_k \mathbb{E} \|\hat{\mathbf{v}}_k - \mathbf{v}_k\|^2, \quad w_1 > \dots > w_K. \quad (5)$$

The total objective is $\mathcal{L} = \mathcal{L}_v + \mathcal{L}_a + \lambda \mathcal{L}_{\text{CoF}}$. Specific weight values are given in the Appendix.

4 Experiments

We evaluate MobileWAM on the ManiSkill-HAB benchmark, ablate every design decision along the evolution path that led to CoF and the mobile MoE, measure deployment latency against imagine-then-execute WAMs, and validate transfer to a real mobile manipulator.

4.1 Setup

Benchmark. ManiSkill-HAB (Shukla et al., 2025) is a GPU-parallelized, low-level-control re-implementation of the Home Assistant Benchmark (Szot et al., 2021) in ManiSkill3 (Tao et al., 2024), featuring a Fetch robot performing home-scale rearrangement. We adopt the seven subtask-object-scene combinations of the *SetTable* suite: picking an apple from the fridge, placing the apple on the table, opening the fridge door, picking a bowl from the counter, placing the bowl on the table, opening the counter drawer, and closing the counter drawer. Success is measured by closed-loop rollout: an episode succeeds only if the subtask’s completion predicate holds at termination. We report mean and standard deviation of success rate (S.R.) over three independent evaluation runs.

Training. For each subtask, we use 1,000 filtered demonstration trajectories (100 held out for validation) from the benchmark’s data-generation pipeline. MobileWAM is trained in a *single* stage with AdamW (learning rate 10^{-5} , cosine schedule, bf16), a global batch size of 256, and full fine-tuning of both experts; the VAE and text encoder stay frozen. Loss weights are $\lambda_v = \lambda_a = 1$ and $\lambda = 0.1$ with depth decay $\mathbf{w} = (0.4, 0.2, 0.1)$, $K=3$. Observations are 384×640 composited RGB frames; no depth, no point clouds, no privileged states. At inference, we denoise actions in 20 flow-matching steps and execute the full 4-step chunk before replanning. Ablations (Tables 3–6) use a reduced 5,000-step budget to expose trends economically, so their absolute numbers are lower. Further implementation details are in the Appendix.

4.2 Comparison with the State of the Art

Table 1 compares MobileWAM against representative imitation-learning and mobile manipulation policies. MobileWAM attains the best mean success rate (73.0%), leads on five of the seven subtasks, and never collapses anywhere. Every baseline, by contrast, drops to single-digit success on at least one subtask. The margins are substantial over the strongest reported baseline AnchorVLA (Lim et al., 2026), which evaluates only six of the seven subtasks, and +17.4 mean points over AC-DiT (Chen et al., 2025b). This is achieved with a strictly leaner recipe: no 3D point-cloud stream (on which DP3 (Ze et al., 2024) also relies, to little avail), no two-stage mobility-pretraining curriculum, and a comparable demonstration budget. The spatio-temporal consistency that other pipelines build by hand through point clouds, auxiliary decoders, and privileged GT segmentation (Tu et al., 2026) is inherited in MobileWAM from video pretraining and sharpened by CoF. Two-dimensional observations alone turn out to support this capability surprisingly well.

Table 1 Main results on ManiSkill-HAB (SetTable). Success rate (% , mean $\pm$ std over three evaluation runs) on the seven subtask combinations. MobileWAM uses RGB-only inputs and a single training stage. **Bold** marks the best and underline the second best per row (i.e., per subtask across methods); “–” denotes not reported. Baselines: ACT (Zhao et al., 2023); DP (Chi et al., 2023); DP3 (Ze et al., 2024); RDT (Liu et al., 2025); AC-DiT (Chen et al., 2025b); AnchorVLA (Lim et al., 2026).

Subtask	ACT	DP	DP3	RDT	AC-DiT	AnchorVLA	MobileWAM
Pick Apple	28.0 $\pm$ 2.2	21.3 $\pm$ 3.3	0.0 $\pm$ 0.0	12.0 $\pm$ 11.3	<u>33.3$\pm$1.9</u>	22.7 $\pm$ 0.9	46.0$\pm$0.8
Place Apple	8.7 $\pm$ 3.3	28.0 $\pm$ 8.0	31.0 $\pm$ 0.8	32.0 $\pm$ 5.7	33.3 $\pm$ 9.4	64.3$\pm$0.8	<u>63.7$\pm$3.2</u>
Open Fridge	2.0 $\pm$ 2.2	7.3 $\pm$ 5.8	0.0 $\pm$ 0.0	82.7 $\pm$ 10.5	<u>90.7$\pm$5.0</u>	88.9 $\pm$ 0.8	99.3$\pm$0.5
Pick Bowl	28.0 $\pm$ 2.4	20.7 $\pm$ 3.3	20.0 $\pm$ 2.4	10.7 $\pm$ 6.8	36.0 $\pm$ 6.5	<u>44.5$\pm$0.8</u>	46.0$\pm$2.6
Place Bowl	13.0 $\pm$ 0.8	69.3$\pm$3.3	32.0 $\pm$ 0.8	18.7 $\pm$ 5.0	17.3 $\pm$ 6.8	63.8 $\pm$ 2.0	<u>64.7$\pm$1.2</u>
Open Drawer	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	44.0 $\pm$ 8.6	<u>81.3$\pm$6.8</u>	–	91.0$\pm$0.8
Close Drawer	85.7 $\pm$ 1.2	55.0 $\pm$ 5.7	68.0 $\pm$ 0.0	100.0$\pm$0.0	<u>97.3$\pm$1.9</u>	100.0$\pm$0.0	100.0$\pm$0.0
Mean	23.6	28.8	21.6	42.9	55.6	<u>64.0</u>	73.0

Table 2 Component ablation. Starting from the plain WAM, adding CoF and then the mobile MoE monotonically improves the mean success rate. Same protocol as Table 1.

Variant	Pick Apple	Place Apple	Open Fridge	Pick Bowl	Place Bowl	Open Drawer	Close Drawer	Mean
WAM only	39.7 $\pm$ 2.1	45.7 $\pm$ 1.5	97.0 $\pm$ 1.0	39.7 $\pm$ 2.1	46.3 $\pm$ 1.5	<u>89.7$\pm$2.1</u>	100.0$\pm$0.0	65.4
+ CoF	44.7 $\pm$ 1.5	52.3 $\pm$ 2.5	98.0 $\pm$ 1.0	44.3 $\pm$ 0.6	56.3 $\pm$ 1.5	87.0 $\pm$ 1.0	100.0$\pm$0.0	<u>68.9</u>
+ mobile MoE	46.0$\pm$0.8	63.7$\pm$3.2	99.3$\pm$0.5	46.0$\pm$2.6	64.7$\pm$1.2	91.0$\pm$0.8	100.0$\pm$0.0	73.0

4.3 Ablation Studies

Does each component pay its way? Table 2 dissects the full model. Adding CoF to the plain WAM lifts the mean success rate from 65.4% to 68.9% (+3.5), with zero inference cost. Gains appear on both pick and place under a moving viewpoint (Pick Apple +5.0, Place Bowl +10.0). Stacking the mobile MoE on top adds another +4.1, with the largest jump on Place Apple (+11.4). That subtask interleaves base repositioning with precise release, the very regime conflict the MoE is built to resolve. Together the two modules compound rather than substitute: foresight densifies temporal supervision while MoE resolves action-space interference.

Parallel or serial? MLP or transformer? CoF was not designed in one stroke; Table 3 retraces the evolution. The first instinct for densifying supervision is *parallel* foresight: extend the prediction horizon and supervise all future chunks side by side, with no chaining. This helps (50.2 $\rightarrow$ 52.3), confirming that more future supervision is more signal. The gain is modest, however, because parallel targets share no causal structure; the model can fit each future independently without learning how one begets the next. The obvious fix is to serialize: let each foresight step condition on its predecessor’s belief. Doing so with the simplest recurrence, an MLP, backfires badly (46.3): the bottleneck is too crude, the chained representation collapses, and the corrupted belief pollutes the shared backbone through gradients. A transformer-based foresight module restores the expressiveness the chain needs and delivers the best result (58.2). The lesson is crisp: *causally chained* supervision beats parallel supervision, but only if the belief state is rich enough to carry a world forward. In short, denser futures help only when the inductive bias matches the causal structure of mobile manipulation.

Where should the belief state come from? Table 4 probes which backbone layers should seed the chain. Uniformly sampling four layers wins decisively; both indiscriminate concatenation (all 30 layers, 37.1) and any single-region selection (first/middle/last four) underperform. We read this as a signal-to-interference trade-off: a sparse, depth-spanning sample covers the geometry-to-semantics spectrum, whereas dumping every layer into the fusion MLP drowns the physical signal and drags 30 layers’ worth of gradients into one auxiliary head. Empirically, four taps strike the best balance between coverage and gradient noise.

Table 3 Parallel vs. serial foresight (reduced 5,000-step budget; decoupled action denoising). Dense future supervision helps only when it is chained causally and the chaining module is expressive enough. **Bold** marks the best and underline the second best per row.

Subtask	WAM only	+ parallel foresight	+ MLP-style CoF	+ Transformer-style CoF
Pick Apple	<u>24.3±1.5</u>	12.7±1.2	8.3±0.6	32.3±1.5
Place Apple	21.7±2.1	36.0±2.0	<u>32.7±1.2</u>	27.3±1.2
Open Fridge	100.0±0.0	100.0±0.0	<u>97.0±1.0</u>	100.0±0.0
Pick Bowl	<u>19.7±2.1</u>	12.7±1.2	<u>12.7±1.2</u>	36.0±2.0
Place Bowl	34.3±1.2	56.7±1.2	33.0±1.0	<u>35.7±2.1</u>
Open Drawer	<u>51.7±0.6</u>	47.7±0.6	40.7±1.5	76.0±2.0
Close Drawer	100.0±0.0	100.0±0.0	100.0±0.0	100.0±0.0
Mean	50.2	<u>52.3</u>	46.3	58.2

Table 4 Which layers feed the chain? Mean S.R. (%) under the reduced budget, decoupled action denoising.

Layer selection	Mean S.R.
All 30 layers	37.1
Uniform 20 layers	42.7
Uniform 12 layers	50.8
Uniform 4 layers	58.2
First and last layer	55.6
Last 4 layers	52.7
Middle 4 layers	50.0
First 4 layers	54.7

How far should foresight reach? Table 5 varies the chain length. Performance is remarkably stable ($K=1$ to 4 within 3.6 points), peaking at $K=3$. This roughly matches the horizon over which the benchmark’s subtask dynamics remain predictable. We fix $K=3$ throughout. Longer chains add little once the near-term causal window is already covered.

One MoE or two action experts? A tempting alternative to our MoE is architectural separation: split the action expert into *two* equal-sized experts, one denoising the locomotion dimensions and one the manipulation dimensions, with all other attention kept identical to ours. Table 6 evaluates three masks between the two: bidirectional, manipulation-attends-locomotion, and the reverse. Every variant falls well short of our three-expert MoE. The reason, we believe, is that whole-body dimensions are not separable streams but facets of one simultaneous motion. A hard split halves the data each expert effectively learns from, severs the shared parameters that consolidate common structure (kinematics, visual grounding), and forces base-arm coordination to squeeze entirely through cross-expert attention. Restricting that channel one-way degrades results further still. Soft routing over a shared expert keeps one coherent trajectory-level representation while letting specialization emerge: separation of concerns without separation of knowledge. Soft MoE thus outperforms hard architectural splits under the same parameter budget.

Deployment efficiency. Table 7 times one full prediction cycle on an NVIDIA A800. Imagine-then-execute WAMs must denoise an entire video before acting. MobileWAM discards both the foresight chain and the video branch at deployment, encodes the current observation once, and denoises actions against the cached backbone features. This yields $5.3\times$ and $8.7\times$ speedups over Motus and LingBot-VA, respectively. The world model pays for itself at training time and then gets out of the way. Decoupled denoising is therefore not only a modeling choice but a practical deployment requirement for mobile platforms.

4.4 Real-Robot Experiments

We fine-tune MobileWAM on teleoperated demonstrations collected with an ARX Lift2 mobile manipulator and evaluate five household tasks of increasing horizon, from single-step drawer opening (T_a), through shelf

Table 5 Foresight chain length K . Mean S.R. (%) under the reduced budget.

K	1	2	3	4
Mean S.R.	55.7	54.6	58.2	56.3

Table 6 One MoE or two action experts? Mean S.R. (%) under the reduced budget. “M→L”: one-way mask where manipulation tokens attend to locomotion tokens; “L→M” is the reverse.

	MoE (ours)	Bi-dir.	M→L	L→M
Mean S.R.	58.2	48.8	46.9	44.6

and drawer pick-and-place variants (T_b – T_d), to a compound open–fetch–deposit–close sequence (T_e) (task overview in Figure 1, top-left; full setup in the Appendix). The suite deliberately stresses viewpoint change, base–arm coordination, and multi-stage causal chaining under the same RGB-only observation interface used in simulation. Table 8 compares against $\pi_{0.5}$ (Black et al., 2025) fine-tuned on the same data. MobileWAM wins on every task. The margin is most telling on the longest-horizon T_e : $\pi_{0.5}$ never succeeds while MobileWAM reaches 15%. This ordering confirms our central claim that the value of causally chained representations grows with the causal depth of the task.

4.5 Qualitative Analysis of Generated Futures

Although video generation is disabled at deployment, it remains a window into what the model has learned under CoF and the mobile MoE. Figure 3(a)–(d) shows the futures predicted by our full model while executing the fridge-picking task; Figure 3(e)–(h) shows the same rollout predicted by the Wan backbone merely fine-tuned on the dataset. The contrast is stark. Our predictions respect the rules of spatial perception: as the base advances, objects scale consistently with distance and retain their silhouettes. The fine-tuned-only baseline violates the near-large–far-small law of perspective: objects fail to grow as the robot approaches, and their shapes warp across frames. This corrupts precisely the visual evolution from which coherent actions must be inferred, and helps explain why denser foresight supervision improves closed-loop success even though video is discarded at test time.

4.6 Failure Analysis and Limitations

Failures concentrate on pick-and-place subtasks. Categorizing all failed episodes: *localization errors* (grasp or release pose misses the tolerance) account for $\sim 40\%$; *collision interference* $\sim 25\%$; *no recovery after an initial failure* $\sim 20\%$; *failure to restore the rest pose after completion* $\sim 10\%$; and *premature drops* $\sim 5\%$. The dominant mode clusters on target poses at the fringe of the robot’s workspace, where success demands near-exact inverse kinematics and slight perturbations render the commanded pose unreachable. Open and close subtasks fail far less often: their geometry is coarser and less sensitive to centimeter-level endpoint error. These modes chart the road ahead: stronger spatio-temporal consistency, finer collision awareness, and closed-loop error recovery.

Table 7 Inference latency for one prediction cycle on the same mobile manipulation task (NVIDIA A800).

Model	Latency (ms) ↓
Motus (Bi et al., 2026)	4950
LingBot-VA (Li et al., 2026)	8126
MobileWAM (ours)	938

Table 8 Real-robot success rates on the ARX Lift2 across five tasks of increasing horizon ($T_a \rightarrow T_e$).

Method	T_a	T_b	T_c	T_d	T_e
$\pi_{0.5}$ (Black et al., 2025)	35%	25%	10%	10%	0%
MobileWAM (ours)	55%	35%	25%	20%	15%

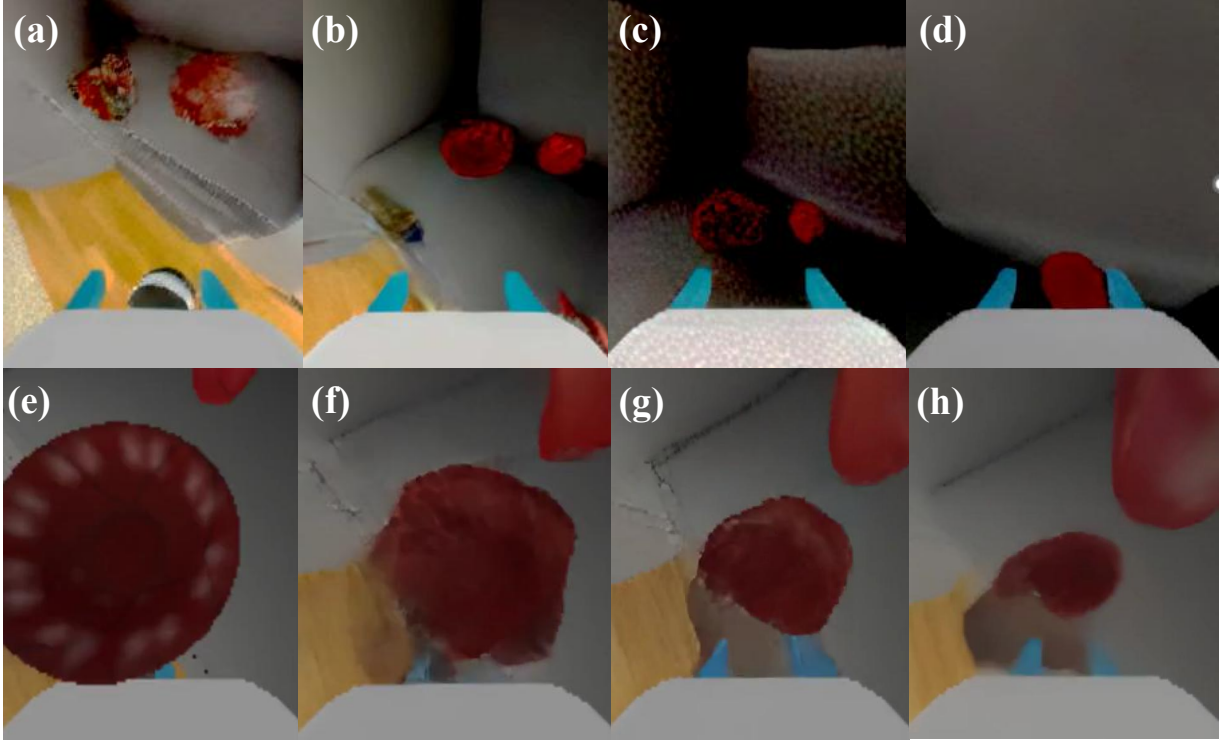


Figure 3 Generated futures during fridge picking. (a)–(d): our full model (CoF + mobile MoE); (e)–(h): the Wan backbone merely fine-tuned on the same data. Ours preserves perspective and shape consistency under base motion; the baseline distorts both.

5 Conclusion

We presented MobileWAM, an early bridge between WAM and mobile manipulation. MobileWAM fuses a pretrained video diffusion transformer with a lightweight action expert via layerwise joint attention, specializes the action pathway with a three-expert (shared, locomotion, manipulation) mixture, and densifies temporal supervision with Chain-of-Foresight, an RNN-style latent dynamics chain that costs nothing at inference. Together, these convert internet-scale motion priors into whole-body control. MobileWAM sets a new state of the art on ManiSkill-HAB with RGB-only inputs and single-stage training, runs at policy-level latency ($5\text{--}8\times$ faster than imagine-then-execute WAMs), and transfers to a real ARX Lift2 with advantages that grow with task horizon.

Looking forward, we see three natural extensions: post-training with reinforcement or interactive corrections, richer geometric grounding beyond RGB, and applying CoF during large-scale WAM pretraining to compound its benefits. We hope MobileWAM persuades the community that mobile manipulation is not merely a harder benchmark for world action models, but their most natural home.

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Appendix

A Implementation Details

Architecture. The world expert is a 30-block video diffusion transformer (hidden width 3072, feed-forward width 14336, 24 attention heads of dimension 128) initialized from a pretrained open-source text-and-image-to-video model; its causal 3D VAE (48 latent channels, $16\times$ spatial and $4\times$ temporal compression) and T5-family text encoder are frozen throughout. The action expert has 30 blocks with hidden width 1024, feed-forward width 4096, and 24 heads of dimension 128, so per-head dimensions match the backbone and joint attention concatenates heads directly. Action chunks ($H=4$, $d_a=13$) are embedded by a linear layer; the diffusion timestep enters through an AdaLN-style modulation. The proprioceptive state ($d_s=18$: goal position, grasp indicator, object pose, and end-effector pose in the base frame) is linearly projected and appended to the text context.

Chain-of-Foresight. The fusion MLP has two layers. Each depth module F_k comprises three transformer blocks mirroring the backbone block design (self-attention with rotary position embeddings under a block-causal mask, cross-attention to language/proprioception/current-observation tokens, feed-forward), operating at the backbone width. All F_k have independent weights initialized cyclically from the final backbone blocks. Foresight targets use a more aggressive noise-schedule shift (10 vs. 5 for the main branches), reflecting the higher uncertainty of deeper futures. Gradients from the chain flow back into the four tapped backbone layers through the fusion MLP; no stop-gradient is applied along the chain.

Mobile MoE. Each action-expert feed-forward layer is replaced by three experts (shared, locomotion, manipulation), each a clone of the pretrained dense layer at initialization. The router is a zero-initialized linear layer on the mean-pooled noisy-action embedding with softmax temperature 1.0; outputs are convex combinations of the three experts. No auxiliary balancing loss is used.

Optimization. AdamW ($\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01), learning rate 10^{-5} with cosine decay and 5% linear warmup, gradient clipping at 1.0, bf16 mixed precision with ZeRO-style sharding. The main models train with a global batch size of 256; reduced-budget ablations train for 5,000 optimizer steps. Loss weights are $\lambda_v=\lambda_a=1$ and $\lambda=0.1$ for CoF, with depth decay $\mathbf{w} = (0.4, 0.2, 0.1)$ and $K=3$. Text and proprioception conditioning are dropped with probability 0.1 during training for classifier-free-guidance-style robustness; images receive color jitter and mild corruption augmentation with probability 0.5.

Decoupled inference. At deployment, the current composited observation is encoded by the frozen VAE; a single backbone forward pass over the clean current-observation tokens populates a per-layer key-value cache; the action chunk is then iteratively denoised for 20 flow-matching steps attending to this cache. The foresight chain and future-video tokens are never instantiated. The robot executes all $H=4$ actions of a chunk before re-observing and replanning.

B Benchmark and Data Details

The seven SetTable combinations pair four subtask types (pick, place, open, close) with articulated fixtures (fridge, kitchen counter) and target objects (apple, bowl). Episodes randomize object placements and robot spawn poses; the robot must coordinate base motion, torso lift, head gaze, and arm control from RGB observations. Demonstrations are generated by the benchmark’s reinforcement-learning-plus-filtering pipeline, which retains trajectories satisfying behavior and safety predicates; we use 1,000 training and 100 validation trajectories per combination. Success rates are computed by closed-loop rollout with the benchmark’s official completion predicates; we report mean and standard deviation over three independent evaluation runs.

The seven subtasks are as follows; representative rollout strips are shown in Figures S1–S4.

1. *Close Drawer*: navigate to the kitchen counter and push its drawer closed.



Figure S1 Simulation task suite — Close/Open Drawer (ManiSkill-HAB). (a) Close Drawer: navigate to the kitchen counter and push its drawer closed. (b) Open Drawer: navigate to the kitchen counter and pull its drawer open.

2. *Open Drawer*: navigate to the kitchen counter and pull its drawer open.
3. *Open Fridge Door*: approach the refrigerator and pull the door open.
4. *Pick Bowl*: grasp the bowl from the kitchen counter top.
5. *Place Apple*: carry the apple and place it at the target position on the dining table.
6. *Place Bowl*: carry the bowl and place it at the target position on the dining table.
7. *Pick Apple*: open (or navigate through the open) fridge door and grasp the apple inside the refrigerator.

Each rollout strip in Figures S1–S4 places one task per column with time top to bottom; within each row the three panels are (left to right) the third-person, head, and wrist views. Full closed-loop rollouts for these subtasks are included in the accompanying supplementary video.



Figure S2 Simulation task suite — Open Fridge Door / Pick Bowl (ManiSkill-HAB). (a) Open Fridge Door: approach the refrigerator and pull the door open. (b) Pick Bowl: grasp the bowl from the kitchen counter top.

C Real-Robot System and Task Suite

Platform. The ARX Lift2 is a wheeled mobile manipulator with a lifting torso, equipped with a head-mounted Intel RealSense 405 RGB camera and a wrist-mounted camera matching our two-view observation format. Because the on-board compute of ARX Lift2 is insufficient to run our $\approx 6.5\text{B}$ -parameter model in real time, we deploy MobileWAM on a remote server equipped with two NVIDIA A800 GPUs (80 GB each). The robot communicates with the server over Wi-Fi: at each replanning step, the current head and wrist camera images are streamed to the server, the server runs one inference cycle ($\approx 938\text{ ms}$), and the resulting action chunk is transmitted back to the robot for execution. For each task we collect teleoperated demonstrations covering randomized object placements and robot start poses, and fine-tune both MobileWAM and the $\pi_{0.5}$ baseline on identical data. Evaluation episodes randomize initial conditions within the training distribution; an episode succeeds only if the full task predicate is satisfied.

Task suite. The five tasks, ordered by increasing horizon and illustrated in Figure S5, are:

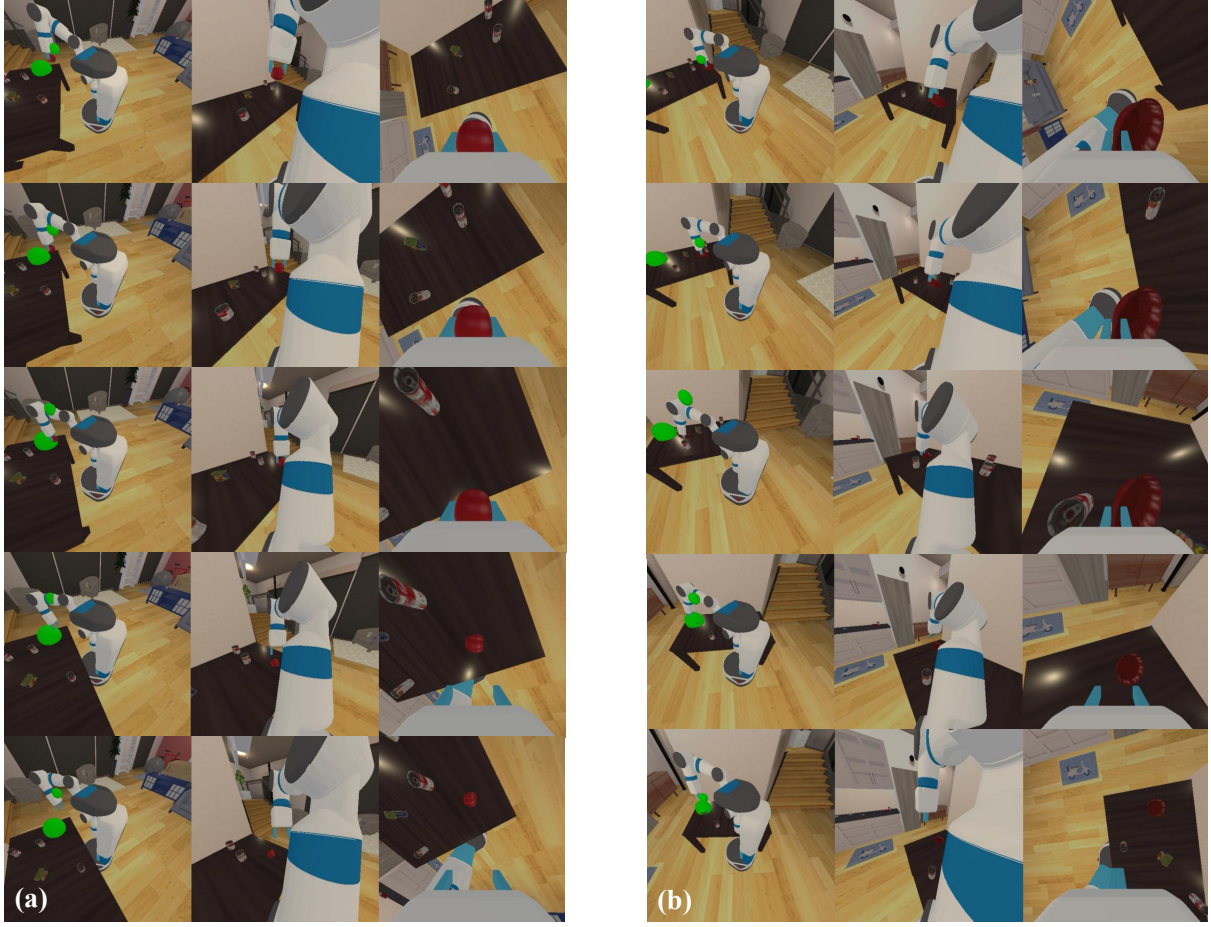


Figure S3 Simulation task suite — Place Apple / Place Bowl (ManiSkill-HAB). (a) Place Apple: carry the apple and place it at the target position on the dining table. (b) Place Bowl: carry the bowl and place it at the target position on the dining table.

- T_a (*Open Drawer*): approach a cabinet and pull its drawer open.
- T_b (*Shelf to Target*): fetch an item from a shelf and place it at a designated location.
- T_c (*Pick to Shelf*): pick an item and place it onto the shelf.
- T_d (*Deposit and Close*): put an item into a drawer, then close the drawer.
- T_e (*Compound*): open the drawer, fetch an item from the shelf, deposit it into the drawer, and close the drawer—a long-horizon composition of T_a – T_d whose later stages causally depend on the earlier ones.

Representative real-robot executions of T_a – T_e can also be viewed in the accompanying supplementary video.

D Additional Qualitative Results

We visualize video predictions from MobileWAM during closed-loop execution in ManiSkill-HAB (Figure S6) and on the ARX Lift2 (Figure S7). In both settings the predicted future frames capture the intended motion trend: object trajectories, end-effector approach directions, and scene-layout changes. This provides a supervisory signal reflected in the learned action distribution even though video generation is discarded at deployment.

E Mobile MoE Expert Routing Analysis

Setup. To understand how the mobile MoE routes action tokens across its three experts (shared, locomotion, manipulation), we visualize per-timestep routing weights alongside locomotion and manipulation speed profiles for all seven SetTable subtasks (Figure S8). For each subtask we compute the mean base speed (norm of the xy -velocity from the wheel-controller action) and the mean arm TCP speed (end-effector displacement relative to the base frame per step), averaged over 1,000 demonstration trajectories. Each timestep is labelled with its dominant modality: *loco-dominant* when base speed exceeds $1.3\times$ arm speed, *manip-dominant* in the reverse case, and *mixed* otherwise, shown as the thin colour strip above each heatmap.

Routing analysis. Across the seven tasks (Figure S8), brighter heatmap rows broadly track the speed-derived phase labels: the locomotion expert rises in loco-dominant segments, the manipulation expert in manip-dominant segments, and the shared expert more often during mixed or transition intervals. Per-timestep weights fluctuate substantially, yet remain relatively balanced (dominant expert typically 0.4–0.6), with no expert collapsing to near-zero—consistent with soft routing, where all experts always contribute to the output.

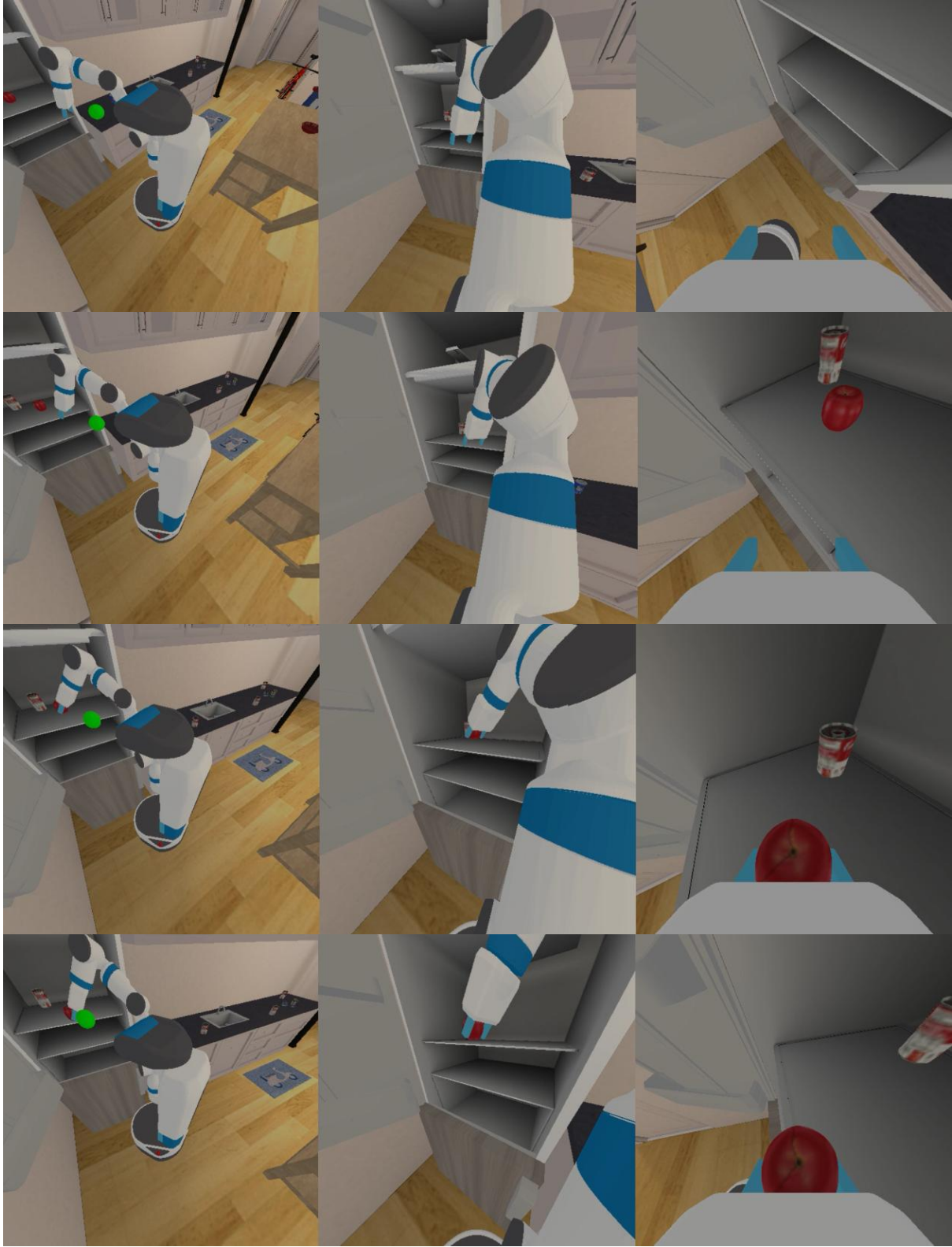


Figure S4 Simulation task suite — Pick Apple (ManiSkill-HAB). Pick Apple: navigate to and open the refrigerator door (if not already open), then grasp the apple inside.

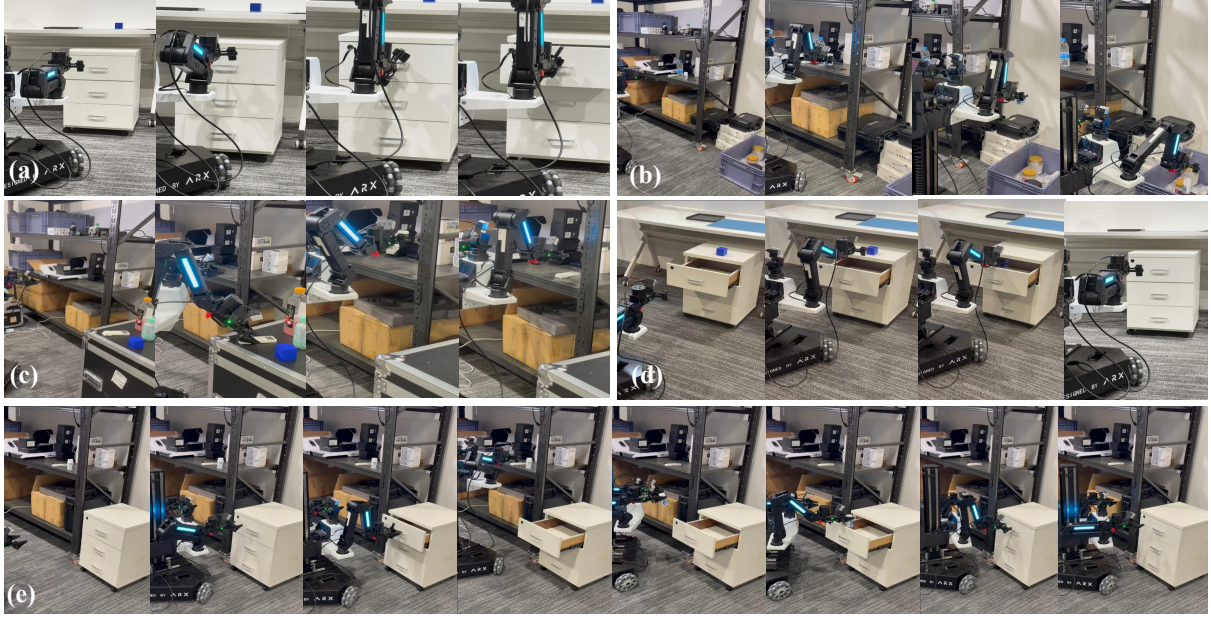


Figure S5 Real-robot task suite on the ARX Lift2. Representative rollout strips for all five tasks T_a – T_e , ordered by increasing horizon. (a) T_a : Open Drawer. (b) T_b : Shelf to Target. (c) T_c : Pick to Shelf. (d) T_d : Deposit and Close. (e) T_e : Compound sequence.



Figure S6 Video generation in simulation (ManiSkill-HAB). Each group shows a partial action segment for one subtask; within each group the left column is the head camera view and the right column is the wrist camera view; the topmost frame of each column is the clean current observation. (a) Open Drawer. (b) Close Fridge Door. (c) Close Drawer.

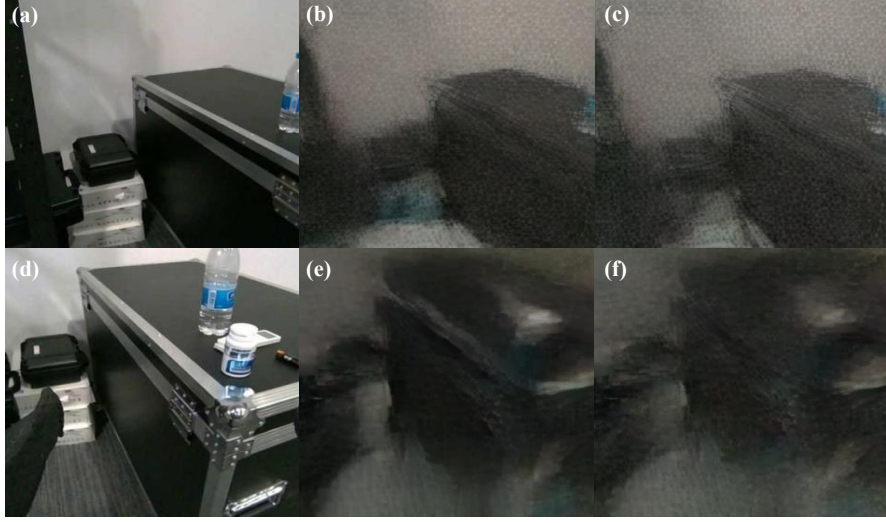


Figure S7 Video generation on the real ARX Lift2 (pick task). A partial action segment during grasping an object from a box. The leftmost frame in each row is the clean current observation. **(a)–(c)** Head camera views at successive timesteps. **(d)–(f)** Corresponding wrist camera views.

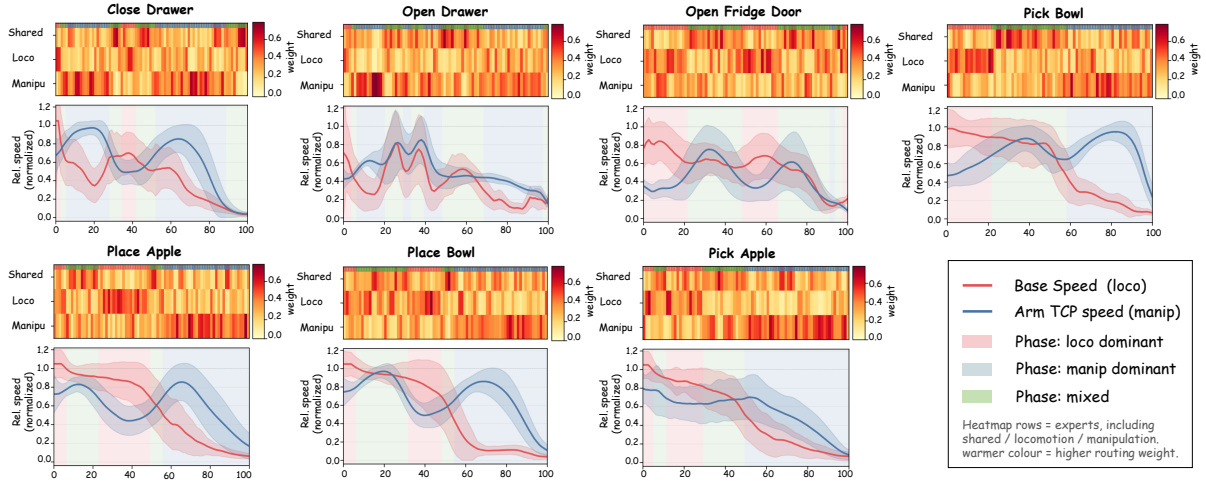


Figure S8 Mobile MoE expert routing weights aligned with locomotion/manipulation speed (SetTable, 7 subtasks). Each panel corresponds to one subtask. *Top*: Routing-weight heatmap; rows are the three experts (shared / loco / manip, top to bottom), and warmer colours indicate higher soft-routing weight at that timestep. The colour strip above the heatmap and the matching background shading on the speed plot mark the inferred dominant phase: **red** = locomotion-dominant, **blue** = manipulation-dominant, **green** = mixed. *Bottom*: Mean normalized base speed (red) and arm TCP speed (blue) ± 1 std across 1,000 demonstrations. Across tasks, the brighter heatmap rows broadly track the phase labels from the speed curves: the locomotion expert rises in loco-dominant segments, the manipulation expert in manip-dominant segments, and the shared expert more often during mixed or transition intervals.